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Linear Search

CODE

```
package main

import "fmt"

func LinearSearch(arr []int, target int) int {
    for i, v := range arr {
        if v == target {
            return i
        }
    }
    return -1
}

func main() {
    arr := []int{1, 2, 3, 4, 5}
    var target int
    fmt.Print("Enter the element to search: ")
    fmt.Scan(&target)
    result := LinearSearch(arr, target)
    if result != -1 {
        fmt.Printf("Element found at index: %d\n", result)
    } else {
        fmt.Println("Element not found in the array.")
    }
}
```

OUTPUT

```
PS E:\VIT_DSA> go run .\LinearSearch.go
Enter the element to search: 3
Element found at index: 2
PS E:\VIT_DSA> go run .\LinearSearch.go
Enter the element to search: 44
Element not found in the array.
PS E:\VIT_DSA> █
```

Binary Search

CODE

```
package main

import "fmt"

func BinarySearch(arr []int, target int) int {
    left, right := 0, len(arr)-1
    for left <= right {
        mid := left + (right-left)/2
        if arr[mid] == target {
            return mid
        } else if arr[mid] < target {
            left = mid + 1
        } else {
            right = mid - 1
        }
    }
    return -1
}

func main() {
    arr := []int{10, 20, 30, 40, 50, 60}

    var target int
    fmt.Print("Enter the element to search: ")
    fmt.Scan(&target)

    result := BinarySearch(arr, target)
    if result != -1 {
        fmt.Printf("Element found at index: %d\n", result)
    } else {
        fmt.Println("Element not found in the array.")
    }
}
```

OUTPUT

```
PS E:\VIT_DSA> go run .\BinarySearch.go
Enter the element to search: 50
Element found at index: 4
PS E:\VIT_DSA> go run .\BinarySearch.go
Enter the element to search: 90
Element not found in the array.
PS E:\VIT_DSA> █
```